Effects of the 100km/h Daytime Speed Limit on Traffic Safety on Dutch Motorways

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1. Introduction

On 16 March 2020 the daytime speed limit on most Dutch motorways was reduced to 100 km/h between 06:00 and 19:00. Before this change, the limit on many of these sections was 120 or 130 km/h, and these higher limits still apply in the evening and at night (Rijkswaterstaat, 2019; Barendrecht & De Jong, 2021).

The measure was introduced to reduce nitrogen deposition in Natura 2000 areas, not to improve road safety. Lower driving speeds are, however, one of the best-established determinants of road safety. According to the Power Model, the number of crashes changes more than proportionally with mean speed, and the more severe the crash, the stronger this effect (Nilsson, 2004; Elvik, 2009). A lower limit is therefore expected to reduce injury crashes relatively more than material-damage crashes. The policy therefore offers a large-scale, real-world setting to study how a speed-limit reduction relates to traffic safety on motorways.

Evaluating this is not straightforward, for two reasons: first, the reduction coincided with the first COVID-19 lockdown, which changed traffic volumes and travel behaviour substantially (Barendrecht & De Jong, 2021). A simple before/after comparison of accident counts would mostly capture this drop in traffic. Second, the open version of the national accident register (hereinafter referred to as: BRON) does not contain the date or time of an accident (Section 4.1), so accidents cannot be split into those that happened under the 100 km/h regime (06:00–19:00) and those that happened under the old limit. This project addresses both issues explicitly: we correct for traffic exposure, compare affected motorway sections with sections whose limit did not change, and use the recorded light condition as a proxy for time of day.

The report proceeds as follows. Section 2 defines the research objective and questions. Section 3 describes the study area, treatment groups and time periods. Section 4 presents the data and preparation procedures. Section 5 explains the analytical strategy. Section 6 reports the available results by research question. Section 7 sets out robustness checks, Section 8 discusses interpretation and limitations and Section 9 presents the conclusion.

2. Research Objective and Research Questions

The objective is to estimate how registered traffic-safety outcomes on Dutch motorway main carriageways changed after the introduction of the 100 km/h daytime speed limit, relative to motorway sections whose posted limit did not change, while accounting for traffic exposure and common year-specific developments.

Main research question:

How did traffic safety on Dutch motorways change after the daytime speed limit was reduced from 120/130 km/h to 100 km/h in March 2020, compared with motorway sections where the speed limit did not change?

Sub-questions:

- SQ1. **Accident frequency**: How did the annual number of registered accidents on affected motorway sections change between the pre-policy period (2018–2019) and the post-policy period (2022–2025), relative to unaffected comparison sections?
- SQ2. **Accident rate** How did the accident rate (accidents per 100 million vehicle-kilometres) change when differences in traffic volume are taken into account?
- SQ3. **Accident severity and type** — How did the number and share of accidents with injuries or fatalities (as opposed to material damage only) change on affected sections, relative to comparison sections, and which crash types (e.g. single-vehicle, rear-end) account for this change?
- SQ4. **Light condition as a proxy for time of day** — Was the change on affected sections larger for accidents that occurred in daylight (largely during hours with the 100 km/h limit) than for accidents in darkness (largely during hours with the old 120/130 km/h limit)?

Together, these questions separate three things that are easily confused: a drop in accidents because there was less traffic, a drop that occurred on all motorways regardless of the policy, and a change that is specific to the sections and hours where the speed limit was actually lowered.

Scope note. Because the data do not support it (Section 4.1), we do not claim to estimate a strict causal effect, and we do not directly compare accidents between 06:00–19:00 and 19:00–06:00. Sub-question 4 is designed as the closest feasible approximation of that comparison.

3. Geographical and Temporal Scope

Geographical scale

The study covers the national motorway network managed by Rijkswaterstaat (road authority type “R” in the National Road Database, NWB), restricted to the **main carriageway** (*hoofdrijbaan*, BST_CODE = HR). Ramps, connector roads and parallel carriageways are excluded, because their speed limits and traffic volumes are not comparable. Exploration of the 2020 and 2023 data shows roughly 22,500 (2020) and 32,400 (2023) registered accidents on these main carriageways, spread over about 55 motorway numbers.

Motorway sections are divided into two groups:

- **Treated sections:** the limit changed from 120 or 130 km/h to 100 km/h during the day (06:00–19:00), with 120/130 km/h remaining in the evening and at night.

- **Comparison sections:** the limit was already 100 km/h (at all hours) before March 2020 and did not change. Sections with lower limits (e.g. 80 km/h ring roads) are excluded from the analysis.

Group sizes. Each INWEVA section is classified using the median speed limit recorded with its accidents before (2018–2019) and after the policy (2021–2023) (Section 4.4).

Group	Length	Accidents per year (2018–2019)	Of which injury / fatal
Treated (120/130 -> 100 km/h)	3,426 km	18,525	631
Comparison (100 km/h before and after)	944 km	9,935	365
Mixed or unclear (too few accidents or other limit combination, excluded)	368 km	671	26
Limit below 100 km/h (excluded)	122 km	1,239	28

The comparison sections are mainly urban ring roads and corridors (A15, A10, A4, A16, A12, A20). They carry about 50% more traffic per kilometre than the treated sections and are more urban in character. This makes it important to compare *changes* within each group rather than levels between the groups.

Temporal scale

The study period is **2018–2025**, determined by the overlapping availability of BRON and of INWEVA traffic volumes on a consistent road network. Because BRON only provides the year of an accident, the temporal resolution of the analysis is **one year**.

Period	Years	Role in the analysis
Pre-policy	2018–2019	Baseline; also used to check whether treated and comparison sections followed parallel trends before the policy.
Transition / COVID-19	2020–2021	Shown in all time series but excluded from the main before/after comparison (policy start mid-March 2020 and strongly disrupted traffic).

Post-policy	2022–2025	Main “after” period.

Figure 1 shows the final classification on the Dutch motorway network.

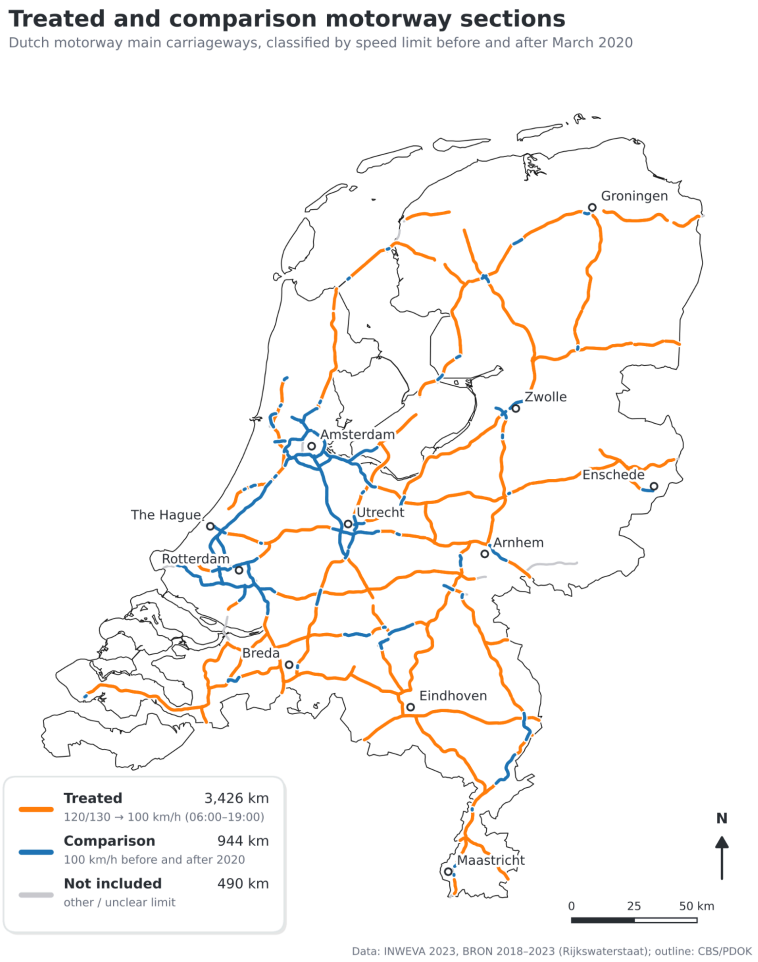


Figure 1: Treated and comparison motorway main carriageways. Treated: daytime limit lowered from 120/130 km/h to 100 km/h in March 2020. Comparison: limit 100 km/h before and after 2020. Classification based on the median recorded speed limit per INWEVA section in 2018–2019 and 2021–2023.
Sources: INWEVA 2023, BRON 2018–2023 (Rijkswaterstaat); outline: CBS/PDOK

Figure 1 shows the location of both groups: 1,901 treated sections (3,426 km) and 759 comparison sections (944 km), while 549 sections (490 km) with a lower, mixed or unclear limit are not included. The comparison group is concentrated around Amsterdam, Rotterdam, The Hague and Utrecht. Because the two groups differ in location and traffic volume, the analysis compares changes within each group rather than levels between them.

4. Data Sources and Preparation

4.1 BRON - Registered Traffic Accidents

The *Bestand geRegistreerde Ongevallen Nederland* (BRON, Rijkswaterstaat) contains registered accidents linked to the NWB road network. Yearly files are available for 2003–2025. It is the main dataset of this project and is used for accident counts, severity, location and light condition.

Data availability check. We inspected the 2020 and 2023 open-data files. To comply with the GDPR (AVG), the open release does not contain variables that could identify individuals: all date and time fields are empty. The table below summarises the fields that matter for this project.

Variable	Meaning	Filled 2020	Filled 2023
JAAR_VKL	Year of accident	100%	100%
DATUM_VKL, MND_NUMMER, WEEKNR, DAG_CODE	Date, month, week, weekday	0%	0%
TIJDSTIP, UUR, DDL_ID	Time, hour, part of day	0%	0%
AP3_CODE	Severity: fatal / injury / material damage only	100%	100%
ANTL_DOD, ANTL_GZH, …	Number of casualties	0%	0%
LGD_ID	Light condition: daylight / darkness / twilight	100%	100%
MAXSNELHD	Speed limit at accident location	93%	91%
AOL_ID	Crash type (rear-end, single-vehicle, side, …)	100%**	100%**

WVK_ID + HECTOMETER	Road section (NWB) and hectometre position	71% / 30%*	74% / 34%*

Percentages over all accidents. For accidents on motorway main carriageways the hectometre position is filled for 93% (2020) and 96% (2023). * Always filled, but the value is "unknown" for about 70% of motorway accidents. Crash type is known for 95% of injury accidents and for only 24% of material-damage accidents.

Consequences for the research design. (1) Accidents cannot be aggregated by month or by hour, so the unit of analysis becomes *road section* × *year*. (2) A direct day/night comparison is not possible; instead, the fully filled light-condition variable is used as a proxy (sub-question 4). (3) Severity is analysed with the three-class AP3_CODE; since fatal accidents on motorways are rare (about 45–55 per year), fatal and injury accidents are combined into one "injury accidents" category. (4) MAXSNELHD follows the daytime limit in force: in 2020, 44% of motorway main-carriageway accidents still carry 120 or 130 km/h, whereas in 2023 this is below 1% overall and below 3% for accidents in darkness, even though the higher night limit applies then. The field is therefore only used to classify sections, using the median recorded limit in 2018–2019 and in 2021–2023, not to tell day from night. (5) Crash type is only analysed for injury accidents, where it is almost always known. A more detailed BRON extract with dates is only available to researchers under a formal data-processing agreement with Rijkswaterstaat, which is not feasible within this course.

4.2 INWEVA — Traffic Intensity on Road Sections

INWEVA (*INtensiteit WEgVAkken*, Rijkswaterstaat, CC-0) provides yearly traffic volumes for every carriageway section (*baanvak*) of the national road network. We downloaded and inspected the releases for 2019–2025. The 2019 release is summarised below; the differences between releases follow after the table.

Content (INWEVA 2019)	**What it contains / what we found**
Sections	8,770 sections, of which 4,394 are motorway main carriageway (HR): about 7,900 km on 60 road numbers.
Location fields	Road number, direction, hectometre from/to, carriageway type, section length in metres.
NwbId	The NWB wegvak IDs that make up each section. This is the same key that BRON uses (WVK_ID), so accidents can be linked directly.

Daily volumes	Per day type (average day of the week, working day, weekend day), for all vehicles and per vehicle class (passenger cars, medium and heavy trucks), with lower and upper bounds.
Period volumes	Day (07–19 h), evening (19–23 h) and night (23–07 h).
Hourly volumes	Separate files with the average volume for each of the 24 hours of the day, per section and vehicle class.
Data quality	For HR sections: 63% measured, 12% derived, 22% traffic model, 3% manual correction (column Bron_AL), plus a reliability score of 1–6.
Year-to-year linking	A table that links each section to its section number in the previous year (folder Netwerkmutaties).

Three release formats. The releases differ in ways that determine how they are used:

Releases	Format and network	Use in this project
2019–2023	Excel files on the VIAS network. Since 2021 each release also contains a *historie* file.	The 2023 historie file gives volumes for 2018–2023 on one and the same network (4,316 HR sections, all with values for every year). This is the main exposure source for 2018–2023, so no year-to-year linking of sections is needed.
2024–2025	New CSV format on the SHiVi network (*verkeersbanen*), produced with a new method by NDW and Rijkswaterstaat.	Linked to the VIAS sections through the column vias_baan_ids (96% of HR sections) and to BRON through nwb_ids (100%).

Link test with BRON. Using NwbId, 93% of the 2020 accidents on motorway main carriageways could be matched to an INWEVA 2019 section, and the road numbers agree in 99.4% of matches. With the 2019 network, the match rate for 2023 accidents is still 87%. The final analysis uses the 2023 network for 2018–2023 and the SHiVi network for 2024–2025, with this approach, 95–97% of motorway main-carriageway accidents per year were linked to an INWEVA section (95.1% in 2018 to 97.2% in 2024).

Function: traffic exposure. For each section and year we compute vehicle-kilometres = average daily volume (all days of the week) × 365 × section length. This is the denominator of the accident rate and the exposure term in the regression model. The hourly volumes add a time dimension that BRON lacks. In 2019, **82% of all vehicle-kilometres on motorway main carriageways were driven between 06:00 and 19:00**, the hours of the 100 km/h limit. The policy therefore applies to the large majority of motorway exposure,

even though accidents cannot be dated. The historie file also shows the size of the COVID-19 shock: the length-weighted average daily volume on main carriageways fell from about 29,500 vehicles in 2019 to 24,100 in 2020 (−18%) and recovered to 28,800 in 2023. Rijkswaterstaat reports a comparable drop of about 17% (Rijkswaterstaat, 2021). This confirms that comparing raw accident counts without exposure correction would be misleading.

4.3 NWB — National Road Database (included in BRON download)

Each BRON download contains the matching version of the NWB road network (wegvakken, hectopunten and road geometry as shapefile). **Function:** linking accidents to motorway number, carriageway type (HR), driving direction and hectometre position, filtering the motorway main carriageway, and drawing maps.

4.4 Speed-limit classification

The speed limit of each section before and after the policy is derived from the limit recorded with its accidents in BRON (MAXSNELHD). For each INWEVA section, the median recorded limit is calculated for 2018–2019 and for 2021–2023. A section is classified as treated if the median was 120 or 130 km/h before and 100 km/h after the policy, and as comparison section if the median was 100 km/h in both periods (Section 3, Figure 1). The traffic decision of 17 December 2019 (Rijkswaterstaat, 2019) describes the policy itself.

4.5 Actual driving speeds (not included)

Historical speed measurements from NDW (Dexter) could show whether actual speeds fell on treated sections. Because access requires an account and the data volume is large, these data were not used in this project. Instead, the speed effects reported in the Rijkswaterstaat monitoring study (Barendrecht & De Jong, 2021) are used as context in the discussion.

5. Analytical Methods

Descriptive comparisons

Annual accident counts are compared between treated and comparison road sections for the period 2018–2025. The main before–after comparison uses 2018–2019 as the pre-policy period and 2022–2025 as the post-policy period. The years 2020–2021 are excluded from this comparison because of the COVID-19 pandemic and are shown separately as a transition period.

Traffic exposure and accident rates

Traffic exposure is measured in vehicle-kilometres (VKM), calculated by multiplying average daily traffic volume by road-section length in kilometres and 365 days:

$$VKM = Average\ Daily\ Traffic \times Road\ Length \times 365$$

Accident rates are expressed as the number of accidents per 100 million vehicle-kilometres:

$$Accident\ Rate = \frac{Accidents}{VKM} \times 100.000.000$$

Traffic data for 2018–2023 are obtained from historical INWEVA data, while data for 2024–2025 are obtained from the newer INWEVA format. This adjustment accounts for differences in traffic volumes between road sections and over time.

Difference-in-Diffferences and event study

A conditional Poisson regression model is used to estimate changes in accident rates. The model includes road-section fixed effects, year fixed effects and the logarithm of vehicle-kilometres as an offset. Standard errors are cluster-robust at the road-section level to account for correlation between observations from the same section over time.

The event-study model includes an interaction between the treated group and each year, with 2019 as the reference year:

$$\log(\mu_{it}) = \log(VKM_{it}) + \alpha_i + \lambda_t + \sum_{k \neq 2019} \beta_k (T_i \times Year_{kt})$$

The main Difference-in-Differences (DiD) model is the same, but replaces the yearly interactions with a single interaction between the treated group and the post-policy period (2022–2025). The exponentiated interaction coefficient (e^β) is interpreted as an incidence-rate ratio (IRR). An IRR below 1 indicates a larger reduction in accident rates on treated roads relative to comparison roads.

The event study shows the yearly differences between the two groups before and after the policy introduction, while the DiD model summarises the overall change in one estimate.

The main identifying assumption is that treated and comparison roads would have followed parallel trends in the absence of the policy. However, only two pre-policy years are available, and the 2018 estimate differs significantly from the 2019 reference year. Therefore, this assumption cannot be confidently established, and the results should not be interpreted as definitive causal effects.

Two sensitivity analyses are also performed: using only 2019 as the pre-policy period and using a stricter comparison group consisting of roads that consistently had a 100 km/h speed limit before the policy.

Injury accidents and crash types

Accidents are classified using the BRON severity variable (AP3_CODE). Fatal (DOD) and injury (LET) accidents are combined into one category, while material-damage-only

accidents (UMS) are analysed separately. Exposure-adjusted Poisson models are used to compare changes in accident rates between these categories.

Crash types are analysed descriptively among injury and fatal accidents, using the AOL_ID variable. The categories include rear-end collisions, single-vehicle accidents, side collisions and collisions with fixed objects. Remaining or unclassified accidents are grouped as other/unknown.

Daylight and darkness analysis
Since BRON does not contain the exact time of an accident, recorded light conditions are used as a proxy for the time of day. Accidents are classified as daylight, darkness or twilight. Twilight is included in the descriptive analysis but excluded from the regression models.

Separate event studies are estimated for daylight and darkness. In addition, a Triple Difference (DDD) model directly tests whether the change in accident rates on treated roads differs between daylight and darkness.

The exponentiated triple interaction coefficient represents the ratio of the daylight and darkness DiD estimates. A value below 1 indicates a larger relative reduction in daylight.

Light-specific exposure is not available, so total vehicle-kilometres are used as exposure for both light conditions. Because the model includes road-section and year fixed effects, this only affects the results if the share of traffic in darkness changed differently on treated and comparison roads. A more important limitation is that light condition does not perfectly correspond to the hours during which the speed limit changed. The results are therefore interpreted as exploratory evidence rather than a direct estimate of the effect during policy hours.

6. Results

This chapter presents the results of the analysis of the 100 km/h daytime speed limit on Dutch motorways. The results are organised according to the four sub-questions. First, the change in the number of registered accidents is examined. Second, accident rates are analysed while accounting for traffic exposure. Third, the analysis focuses on accident severity and crash types. Finally, the relationship between accident rates and light conditions is investigated.

The main comparison is between the pre-policy period (2018–2019) and the post-policy period (2022–2025). The years 2020 and 2021 are included in the figures to show developments during the transition period, but are excluded from the main before-and-after comparison. This is important because the policy was introduced in March 2020, at the same time as the COVID-19 pandemic, which substantially affected traffic volumes and travel behaviour.

6.1 SQ1 — How did the number of registered accidents change?

Figure 2 presents the annual number of registered accidents on treated and comparison motorway sections between 2018 and 2025. Treated sections are roads where the speed limit

changed from 120 or 130 km/h to 100 km/h during daytime. Comparison sections are roads where the speed limit was already 100 km/h and did not change.

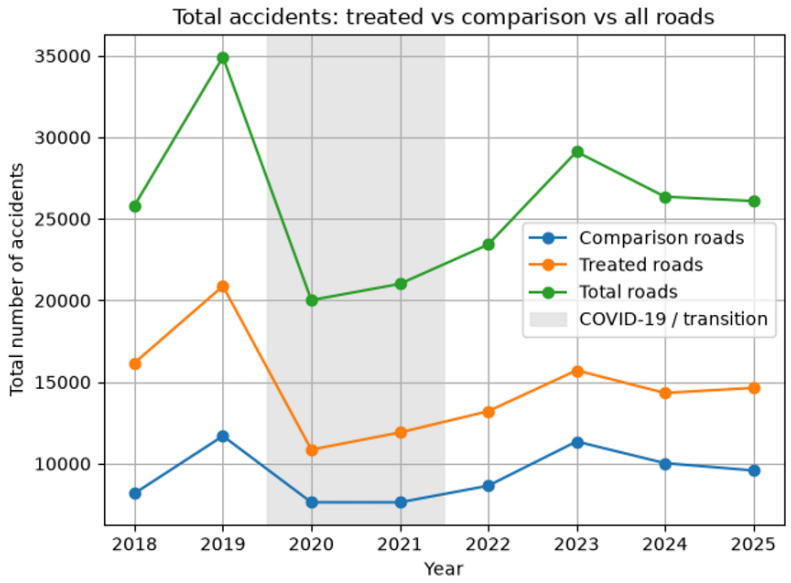


Figure 2. Annual registered accidents on treated and comparison roads, 2018–2025.

The descriptive results show a clear difference between the two road groups. On treated roads, the mean annual number of registered accidents decreased from 18,525 in the pre-policy period to 14,474.75 in the post-policy period. This represents a decrease of 4,050.25 accidents per year, or 21.86%.

On comparison roads, the mean annual number of registered accidents changed only slightly. The annual average decreased from 9,934.5 before the policy to 9,902.5 after the policy, corresponding to a reduction of 32 accidents per year, or 0.32%.

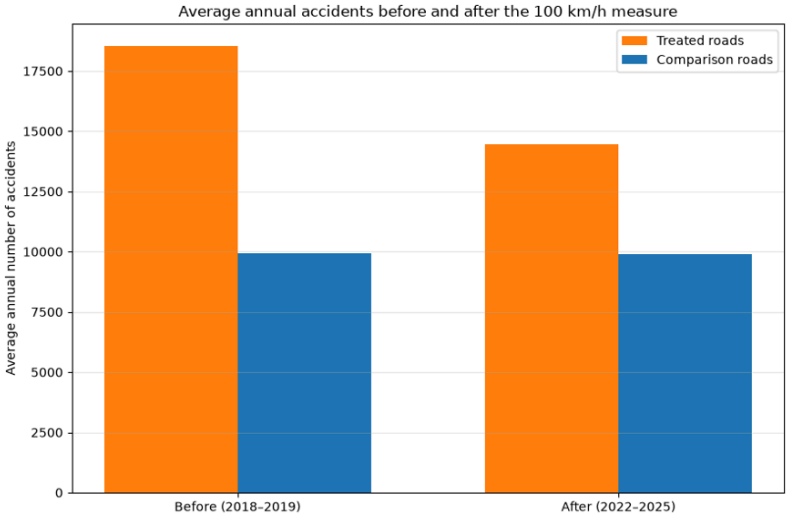


Figure 3. Mean annual number of registered accidents before (2018–2019) and after (2022–2025) the measure.

Table 1. Changes in the mean annual number of registered accidents between the pre-policy and post-policy periods.

Road group	Pre-policy annual mean	Post-policy annual mean	Absolute change	Change (%)
Treated	18,525	14,474.75	−4,050.25	−21.86%
Comparison	9,934.5	9,902.5	−32	−0.32%

These descriptive results indicate that registered accident counts decreased considerably more on treated roads than on comparison roads. The difference between the two groups is consistent with a more favourable development in accident frequency on roads affected by the daytime speed-limit reduction.

However, accident counts alone do not provide a complete picture of traffic safety. The number of accidents is influenced not only by the safety of individual road users, but also by the amount of traffic using the roads. A reduction in traffic can result in fewer accidents even if the risk per kilometre driven remains unchanged. This is particularly relevant for 2020, when traffic volumes decreased substantially due to the COVID-19 pandemic.

Therefore, the results of SQ1 should be interpreted as a comparison of accident frequency rather than accident risk. SQ2 addresses this limitation by taking traffic exposure into account.

6.2 SQ2 — How did accident rates change when accounting for traffic volume?

To account for differences in traffic exposure, accident rates were calculated per 100 million vehicle-kilometres. Vehicle-kilometres combine the amount of traffic with the length of the road sections and provide an indication of how much traffic exposure occurred during each year.

Figure 4 presents the annual accident rates on treated and comparison roads. Figure 5 shows the corresponding traffic-volume trends, with 2019 used as the reference year (index = 100).

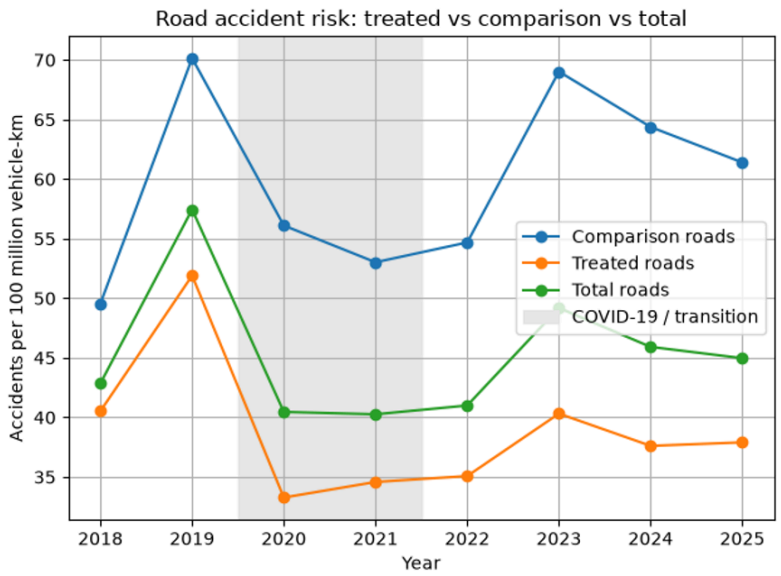


Figure 4. Registered accidents per 100 million vehicle-kilometres on treated and comparison motorway sections, 2018-2025.

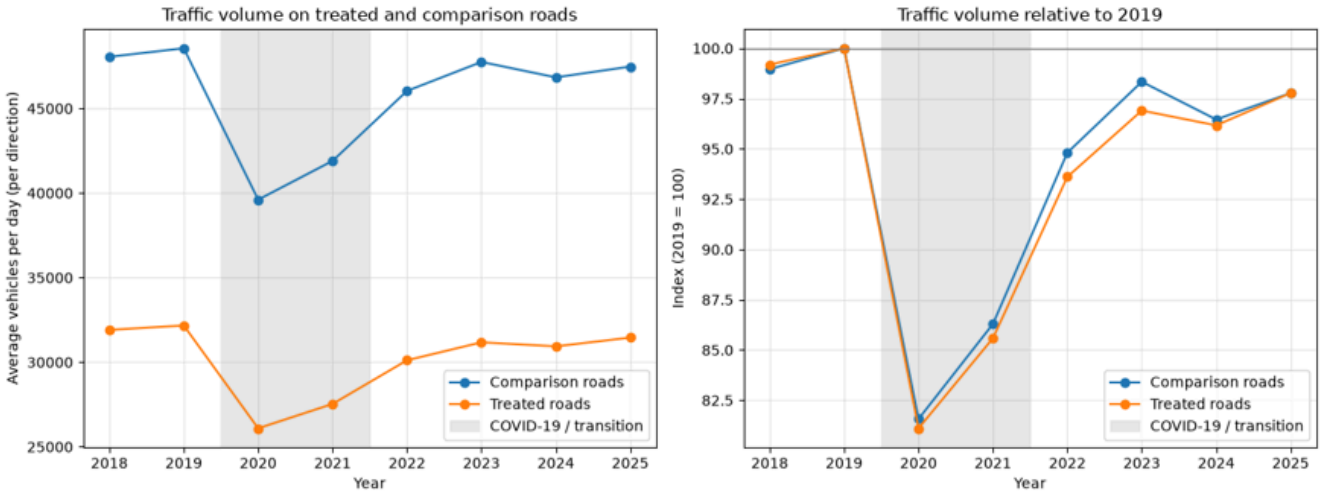


Figure 5. Traffic-volume index for treated and comparison motorway sections, with 2019 = 100.

The traffic-volume trends demonstrate why it is important to account for exposure. In 2020, the traffic-volume index decreased to 81.1 on treated roads and 81.6 on comparison roads, compared with 100 in 2019. This indicates that traffic intensity fell by approximately 19% in both groups. Traffic volumes subsequently recovered, although the index remained below the 2019 level in several years.

By using vehicle-kilometres as an exposure measure, the regression analysis accounts for differences in the amount of traffic using each road section. This makes it possible to compare changes in accident rates rather than relying only on changes in accident counts.

A conditional Poisson Difference-in-Differences (DiD) regression was used to estimate whether accident rates changed differently on treated and comparison roads. The model includes road-section and year effects, with the logarithm of vehicle-kilometres included as an offset. Standard errors are clustered by road section to account for potential correlation between observations from the same road section.

The main model produced an incidence-rate ratio (IRR) of 0.783 (95% confidence interval: 0.753–0.815; $p < 0.001$). An IRR below 1 indicates a more favourable relative change in the treated group. The estimate corresponds to a relative change in accident rates that was 21.7% lower on treated roads than on comparison roads in the post-policy period.

Table 2. Difference-in-differences results and alternative specifications (conditional Poisson regression).

Model specification	IRR	95% confidence interval	p-value	Relative change
Main model	0.783	0.753–0.815	<0.001	−21.7%
2019-only baseline	0.815	0.778–0.853	<0.001	−18.5%
Stricter comparison group	0.767	0.730–0.805	<0.001	−23.3%

The two alternative specifications were estimated to assess whether the main result depends on the selected baseline period or the definition of the comparison group.

The first alternative model uses only 2019 as the pre-policy baseline. It produces an IRR of 0.815 (95% CI: 0.778–0.853; $p < 0.001$), corresponding to a 18.5% more favourable relative change in accident rates on treated roads compared with comparison roads. This estimate is somewhat smaller than the main model estimate of 21.7%, but the direction of the result remains the same.

The second alternative model uses a stricter comparison group. It produces an IRR of 0.767 (95% CI: 0.730–0.805; $p < 0.001$), corresponding to a 23.3% more favourable relative change on treated roads. This estimate is slightly larger than the main model estimate.

All three specifications produce IRRs below 1, and all estimates are statistically significant at the 5% level. The confidence intervals also exclude 1 in each model. The consistency in the direction and approximate magnitude of the estimates suggests that the main finding is relatively robust to these alternative specifications. However, these sensitivity analyses do not resolve possible differences in pre-policy trends between the treated and comparison groups.

To examine how the difference between treated and comparison roads developed over time, an event-study analysis was conducted. The year 2019 was selected as the reference year. This makes it possible to compare the relative accident-rate development in each year with the difference observed in 2019.

Figure 6 presents the estimated incidence-rate ratios for each year, together with their 95% confidence intervals.

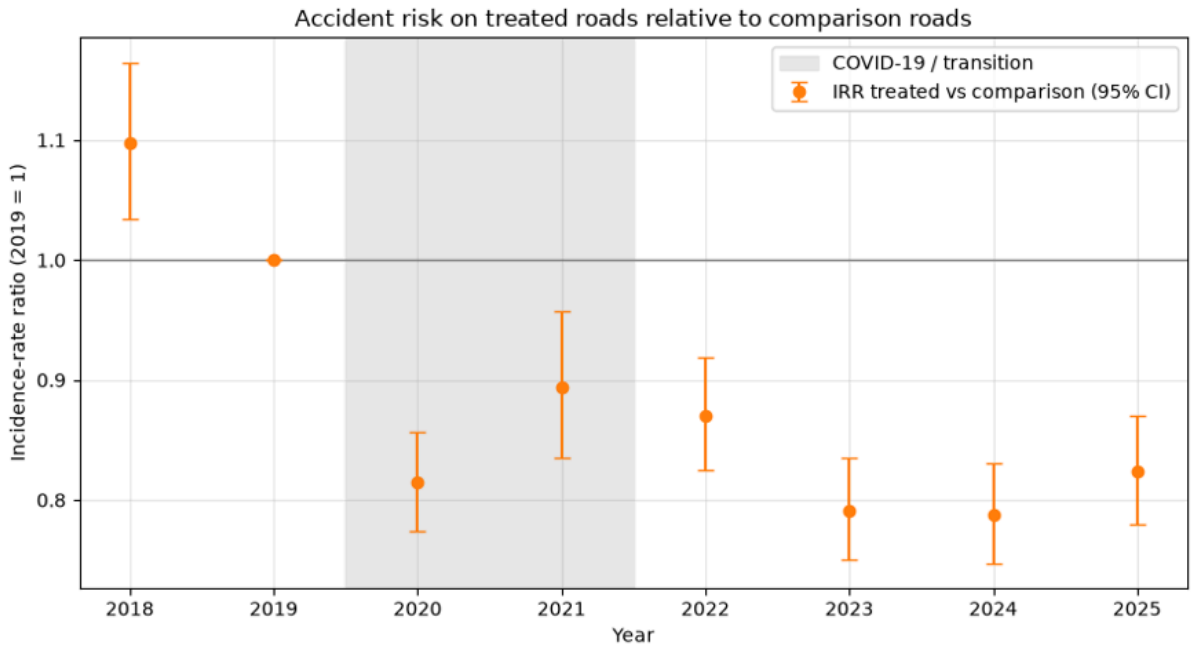


Figure 6. Event-study incidence-rate ratios relative to 2019, with 95% confidence intervals.

The event-study results show that the relative accident-rate estimates were below 1 in all post-policy years. The estimated IRRs were 0.871 in 2022, 0.791 in 2023, 0.788 in 2024 and 0.824 in 2025. These estimates indicate that accident rates on treated roads developed more favourably relative to comparison roads during the post-policy period.

The largest estimated difference occurred in 2024, with an IRR of 0.788. This corresponds to a 21.2% more favourable relative change compared with the reference year. In 2025, the estimate was 0.824, corresponding to a 17.6% more favourable relative change. Although the estimates vary somewhat between years, they consistently remain below 1.

However, the pre-policy estimate for 2018 requires particular attention. The estimated IRR was 1.097 (95% CI: 1.034–1.165). Since the confidence interval does not include 1, this indicates a statistically significant difference between treated and comparison roads already before the policy was introduced.

This finding is important because the difference-in-differences approach relies on the parallel-trends assumption. This assumption means that, in the absence of the policy, accident rates on treated and comparison roads would have followed similar trends over time. The significant estimate for 2018 suggests that the two groups may already have developed differently before March 2020.

The event-study analysis therefore provides two important findings. First, the post-policy estimates consistently indicate a more favourable relative development in accident rates on treated roads. Second, the significant pre-policy estimate raises doubts about whether the parallel-trends assumption is fully satisfied. Since only two pre-policy years are available, it is not possible to establish the pre-policy trend with high certainty.

Overall, the results for SQ2 indicate that accident rates developed more favourably on treated motorway sections than on comparison sections after the introduction of the 100 km/h daytime speed limit. The main difference-in-differences model estimates a 21.7% more favourable relative change in accident rates on treated roads. This result is statistically significant and remains similar across two alternative model specifications.

The event-study analysis also shows lower relative accident-rate estimates in every post-policy year. However, the statistically significant difference in 2018 indicates that the treated and comparison groups may already have followed different trends before the policy was introduced.

Therefore, the results provide evidence of an association between the introduction of the daytime speed limit and a more favourable development in accident rates on treated roads. Nevertheless, the possible violation of the parallel-trends assumption means that the estimated 21.7% difference cannot be interpreted as definitive proof of a causal policy effect.

6.3 SQ3 — Did injury accidents and crash types change differently?

To examine whether the severity of registered accidents changed after the introduction of the daytime speed limit, accidents were divided into three categories: material-damage-only accidents, injury accidents and fatal accidents. Figure 7 presents the annual distribution of these accident categories on treated and comparison roads between 2018 and 2025. The percentages above the bars show the combined share of injury and fatal accidents.

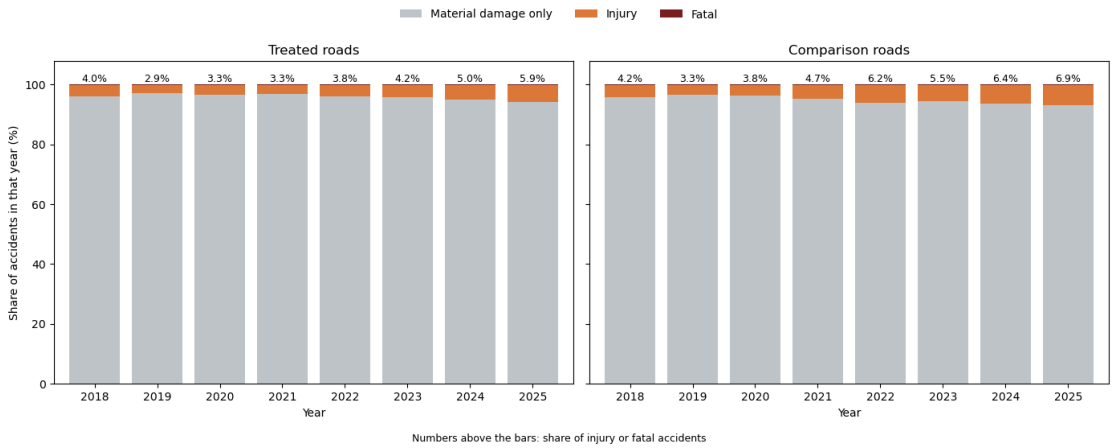


Figure 7. Annual distribution of registered accidents by severity on treated and comparison roads, 2018–2025. Each bar represents 100% of registered accidents in that year. Labels above the bars indicate the combined percentage of injury and fatal accidents.

On treated roads, the combined share of injury and fatal accidents decreased from 4.0% in 2018 to 2.9% in 2019. After the introduction of the policy, this share generally increased, reaching 5.9% in 2025. A similar increase can be observed on comparison roads, where the share increased from 4.2% in 2018 to 6.9% in 2025. This indicates that the proportion of more severe accidents increased in both road groups during the later years of the study period.

However, these percentages represent the distribution of registered accidents, not the risk of being involved in an injury accident. An increase in the injury and fatal accident share does not necessarily mean that the absolute number or rate of injury accidents increased. It may also reflect a decrease in the number of material-damage-only accidents. Therefore, the figure alone cannot establish whether the policy affected injury accident rates.

To account for differences in traffic exposure, Figure 8 compares the average injury and fatal accident rates before and after the policy. The rates are expressed as the number of injury and fatal accidents per 100 million vehicle-kilometres.

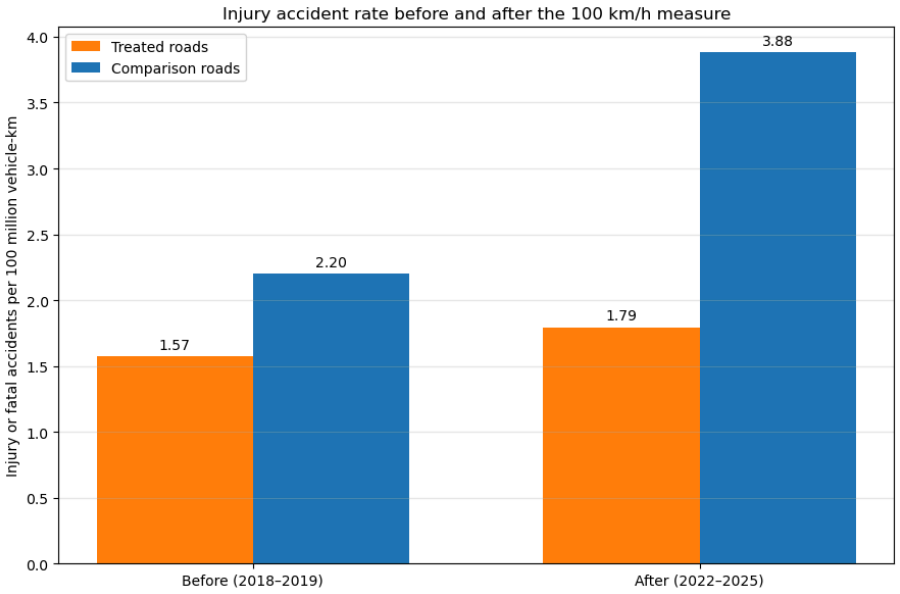


Figure 8. Average injury and fatal accidents per 100 million vehicle-kilometres on treated and comparison roads, before (2018–2019) and after (2022–2025) the policy.

On treated roads, the injury accident rate increased from 1.57 before the policy to 1.79 after the policy. This represents an increase of approximately 14%. On comparison roads, the rate increased from 2.20 to 3.88, which corresponds to an increase of approximately 76%. Although the injury accident rate increased in both groups, the increase was considerably smaller on treated roads. This suggests that injury accident rates developed more favourably on treated roads than on comparison roads during the study period.

Figure 9 examines whether the distribution of crash types among injury and fatal accidents changed after the introduction of the policy. The figure compares the percentage of injury and fatal accidents belonging to each crash type before and after the policy, separately for treated and comparison roads.

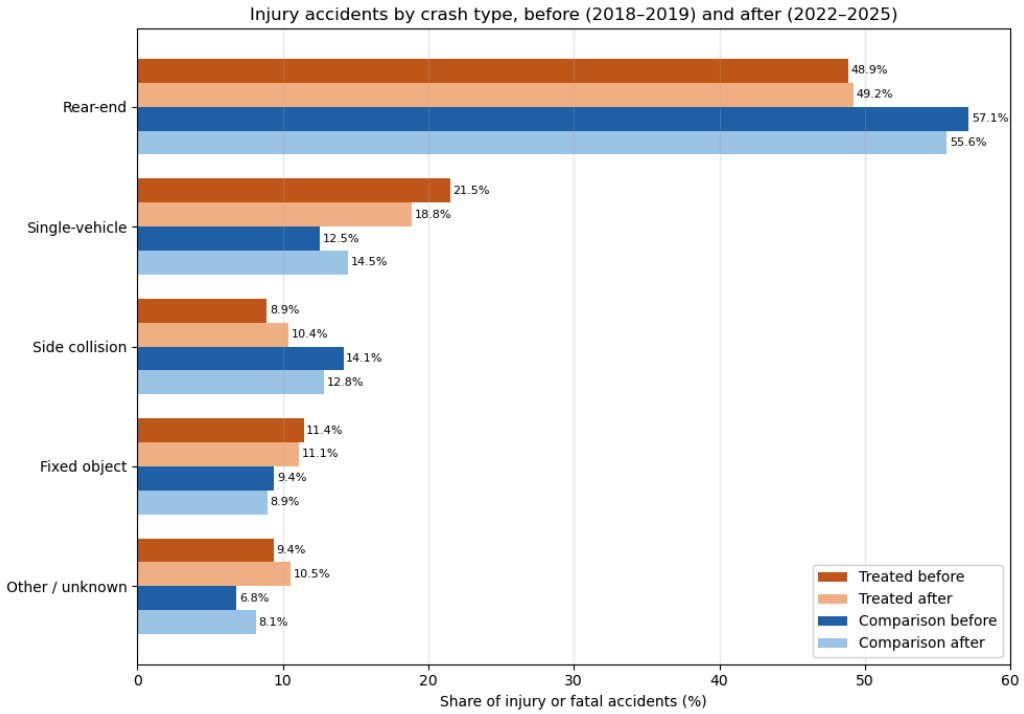


Figure 9. Distribution of crash types among injury and fatal accidents on treated and comparison roads, before (2018–2019) and after (2022–2025) the policy.

Rear-end collisions were the most common crash type in both road groups. On treated roads, their share remained almost unchanged, increasing from 48.9% before the policy to 49.2% after the policy. On comparison roads, the share decreased from 57.1% to 55.6%. The share of single-vehicle accidents decreased on treated roads from 21.5% to 18.8%, while it increased on comparison roads from 12.5% to 14.5%. Side collisions increased slightly on treated roads, from 8.9% to 10.4%, but decreased on comparison roads, from 14.1% to 12.8%.

These results describe changes in the composition of injury and fatal accidents, rather than the exposure-adjusted risk of individual crash types.

To assess whether the differences in accident severity were statistically significant while accounting for traffic exposure, conditional Poisson regression models were estimated. Table 3 presents the estimated incidence-rate ratios (IRRs), 95% confidence intervals and p-values for all accidents, material-damage-only accidents, and injury and fatal accidents.

Table 3. Conditional Poisson regression results for accident rates by severity, including incidence-rate ratios (IRRs), 95% confidence intervals and p-values.

	IRR	95% CI	p-value	Change
All accidents (DiD)	0.783	0.753-0.815	<0.001	-21.7%
Effect: material damage only	0.793	0.762-0.825	<0.001	-20.7%
Effect: injury or fatal	0.650	0.574-0.735	<0.001	-35.0%

Difference: injury or fatal vs material damage only (ratio IRRs)	0.819	0.726-0.925	0.001	-18.1%

The main model for all accidents produced an IRR of 0.783 (95% CI: 0.753–0.815; $p < 0.001$), corresponding to a 21.7% more favourable relative change in accident rates on treated roads compared with comparison roads. For material-damage-only accidents, the estimated IRR was 0.793 (95% CI: 0.762–0.825; $p < 0.001$), representing a relative reduction of 20.7%.

The estimated effect was larger for injury and fatal accidents. The corresponding IRR was 0.650 (95% CI: 0.574–0.735; $p < 0.001$), indicating a 35.0% more favourable relative change on treated roads compared with comparison roads. The ratio of the injury/fatal IRR to the material-damage-only IRR was 0.819 (95% CI: 0.726–0.925; $p = 0.001$). Since this confidence interval excludes 1, the difference between the two estimated effects is statistically significant. In other words, the IRR for injury and fatal accidents was 18.1% lower than the IRR for material-damage-only accidents.

Overall, the descriptive and regression results suggest that injury and fatal accident rates developed more favourably on treated roads than on comparison roads. The regression results further indicate that this relative improvement was significantly larger for injury and fatal accidents than for material-damage-only accidents. However, the results should not be interpreted as definitive proof of a causal policy effect, particularly given the possible violation of the parallel-trends assumption identified in SQ2.

6.4 SQ4 — Were changes greater in daylight than in darkness?

To investigate whether the changes in accident rates differed between daylight and darkness, accidents were classified according to their recorded light condition. Since the open BRON dataset does not contain the exact date and time of each accident, light condition is used as a proxy for the time of day. Daylight largely overlaps with the hours during which the 100 km/h daytime speed limit applied, while darkness largely overlaps with the hours during which the previous speed limit remained in place. Twilight is treated as a separate category because it does not clearly correspond to either period. Figure 10 presents the annual distribution of registered accidents across daylight, darkness and twilight on treated and comparison roads between 2018 and 2025.

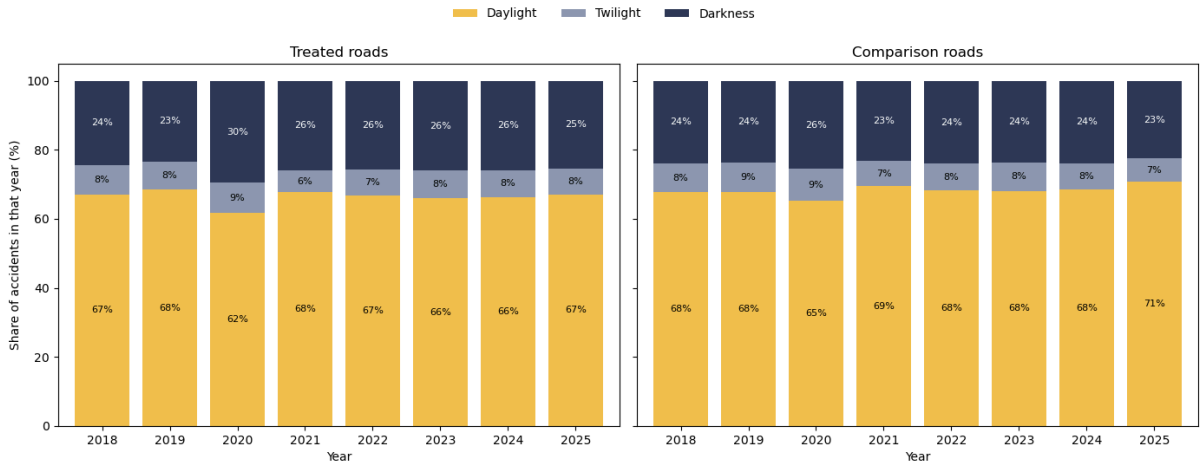


Figure 10. Annual distribution of registered accidents by light condition on treated and comparison roads, 2018–2025.

Figure 10 shows how registered accidents were distributed across daylight, darkness and twilight during the study period. The percentages indicate the share of accidents recorded under each light condition within each road group and year. The figure provides an overview of possible changes in the distribution of accidents over time. However, these percentages do not directly measure accident risk. A change in the share of daylight accidents may also result from changes in the number of accidents during darkness or twilight. Therefore, the figure alone cannot establish whether accident risk decreased more during daylight.

To examine differences in accident risk more directly, an event-study analysis was conducted separately for daylight and darkness. The analysis compares changes on treated roads with changes on comparison roads, while accounting for traffic exposure and differences between road sections and years. The year 2019 is used as the reference year. Figure 11 presents the estimated incidence-rate ratios (IRRs) for daylight and darkness, together with their 95% confidence intervals.

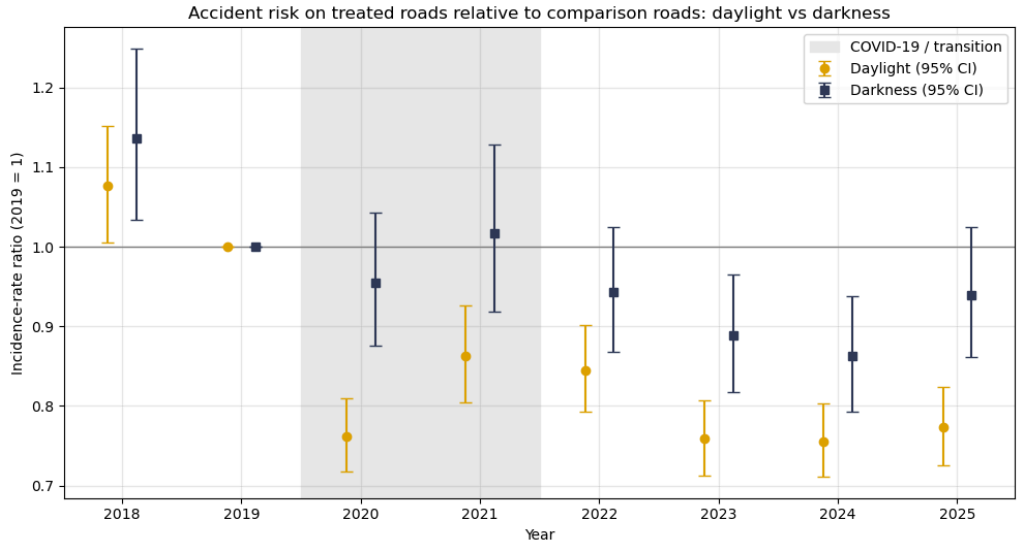


Figure 11. Event-study estimates of accident rates in daylight and darkness on treated roads relative to comparison roads, using 2019 as the reference year. Error bars represent 95% confidence intervals.

In Figure 11, an IRR below 1 indicates that the accident rate was lower relative to the comparison roads than in the reference year. An IRR above 1 indicates a higher relative accident rate. The separate estimates for daylight and darkness show how accident rates developed under the two light conditions. The post-policy estimates should be compared to assess whether the relative decline was larger during daylight than during darkness.

However, comparing the separate daylight and darkness estimates is not sufficient to establish whether the difference between them is statistically significant. Therefore, a triple-difference (DDD) regression was conducted. This model includes the interaction between treated roads, the post-policy period and daylight. It also accounts for traffic exposure and differences between road sections and years. The DDD interaction directly tests whether the change in accident rates on treated roads differed between daylight and darkness, relative to the comparison roads.

Table 4. Triple-difference regression results for daylight and darkness, including the interaction estimate, incidence-rate ratio, 95% confidence interval and p-value.

	IRR	95% CI	p-value	Change
Effect: darkness	0.856	0.806-0.909	<0.001	-14.4%
Effect: daylight	0.755	0.722-0.791	<0.001	-24.5%
Difference: daylight vs darkness (ratio of IRRs)	0.882	0.826-0.943	<0.001	-11.8%

Table 4 presents the results of the difference-in-difference-in-differences (DDD) regression. The estimated IRR for darkness is 0.856, corresponding to a 14.4% more favourable relative change in accident rates on treated roads compared with comparison roads. For daylight, the estimated IRR is 0.755, corresponding to a 24.5% more favourable relative change.

The ratio of IRRs for daylight compared with darkness is 0.882 (95% CI: 0.826–0.943; $p < 0.001$). This indicates that the relative change was 11.8% more favourable during daylight than during darkness. Since the confidence interval does not include 1 and the p-value is below 0.05, this difference is statistically significant. Therefore, the model provides statistical evidence that the relative development in accident rates differed between daylight and darkness.

Overall, SQ4 examines whether accident rates changed more favourably during daylight, when the 100 km/h speed limit applied, than during darkness. The event-study provides a visual comparison of the yearly changes, while the DDD regression directly tests whether the difference between daylight and darkness is statistically significant.

7. Robustness Checks

To check the robustness and validity of the results a sensitivity analysis was performed. Seven alternative approaches were used to perform the regression and calculate the IRR, CI, p-value and changes in accident rate. In Table 5, the results of the seven sensitivity analyses alternatives can be seen compared to the base case used in this paper. In alternative 1 through 4, several combinations of before and after years are considered. In alternative 5, stricter comparison groups are considered and in alternative 6 and 7, only the highest and lowest, respectively, accident rate motorway sections are considered. Note that alternative 1 and 5 have already been discussed in detail in Chapter 6.2.

Table 5. Regression results for alternative method changes in sensitivity analysis.

	IRR	95% CI	p-value	Change
Base case	0.783	0.753 – 0.815	<0.001	-21.7%
Alternative 1: 2019 vs 2022-2025	0.815	0.778 – 0.853	<0.001	-18.5%
Alternative 2: 2018-2019 vs 2022-2023	0.793	0.760 – 0.829	<0.001	-20.7%
Alternative 3: 2018-2019 vs 2024-2025	0.774	0.741 – 0.808	<0.001	-22.6%
Alternative 4: 2018-2019 vs 2022	0.835	0.796 – 0.877	<0.001	-16.5%
Alternative 5: Stricter comparison group	0.767	0.730 – 0.805	<0.001	-23.3%
Alternative 6: Highest 10% accident rates	0.689	0.621 – 0.766	<0.001	-31.1%
Alternative 7: Lowest 10% accident rates	0.968	0.804 – 1.165	0.730	-3.2%

The results are remarkably similar for different alternatives and show the result in the base case of a 21.7% accident drop (IRR: 0.783, CI: 95%, p<0.001) are relatively robust. Additionally, the year where only 2024 and 2025 are considered has the lowest IRR in alternative 1-4, aligning with the results described in Chapter 6.2. Interestingly, the biggest deviation from the base case is found in alternative 6 and - 7. The highest 10% accident rate motorway sections show a large (31,1%) decrease in accident rate, whereas the lowest 10% accident rate motorway sections show no significant (p=0.730, change= -3.2%) change. This implies that the accident rate change in the base case is carried by the change on more dangerous roads. However, this conclusion must be made cautiously due to the insignificance of alternative 7. It is suggested to perform a formal statistical test to find a significant indication of between-group differences in accident rate change.

8. Discussion

8.1 Synthesis and interpretation

Taken together, the results suggest that traffic safety developed more favourably on treated motorway sections than on comparison sections following the introduction of the 100 km/h

daytime speed limit. This pattern can be observed across all four sub-questions, although the findings differ in terms of accident frequency, accident rates, severity and light conditions.

For SQ1, the average annual number of registered accidents decreased by 21.9% on treated roads, compared with only 0.3% on comparison roads. However, accident counts alone do not provide a complete picture of traffic safety, as they are also influenced by traffic volumes. SQ2 therefore accounted for traffic exposure using vehicle-kilometres. The main Difference-in-Differences model showed a 21.7% more favourable relative change in accident rates on treated roads. This result was relatively stable across alternative model specifications. Nevertheless, the event-study showed a significant difference between the groups already in 2018, weakening the parallel-trends assumption and limiting the causal interpretation of the results.

SQ3 examined the relationship between the speed limit reduction and accident severity. The regression results showed a larger relative reduction in injury and fatal accident rates (35.0%) than in material-damage-only accident rates (20.7%). This difference was statistically significant and consistent with the Power Model, which suggests that changes in speed have a stronger effect on more severe accidents (Nilsson, 2004; Elvik, 2009). However, injury and fatal accident rates increased in absolute terms on both road groups, meaning that the more favourable development on treated roads represents a relative improvement.

SQ4 examined differences between daylight and darkness. The DDD model showed a more favourable relative change during daylight (24.5%) than during darkness (14.4%), with a statistically significant difference between the two estimates. This is consistent with the expectation that the speed limit reduction would have a stronger relationship with traffic safety during the hours when the lower limit applied. However, light condition is only an imperfect proxy for the hours of the speed limit: winter mornings and evenings are dark but fall within the 06:00–19:00 period. This may also explain why accident rates in darkness decreased as well (–14.4%).

Overall, the findings are consistent with the theoretical expectations of the Power Model and with Rijkswaterstaat's monitoring of the speed limit reduction (Barendrecht & De Jong, 2021). By examining registered accidents over a longer period and comparing treated roads with roads where the speed limit remained unchanged, our study adds to these findings. The results show a more favourable relative development in accident frequency and exposure-adjusted accident rates on treated roads, particularly for injury and fatal accidents and for accidents recorded in daylight. This suggests that the introduction of the 100 km/h daytime speed limit was associated with improved traffic safety on affected motorway sections. However, differences in pre-policy trends and limitations in the available exposure data mean that the results should be interpreted as evidence of an association rather than definitive proof of a causal effect.

8.2 Limitations

This study has several limitations that should be considered when interpreting the results.

First, the open BRON dataset does not contain the exact date or time of accidents. Therefore, the analysis is based on yearly totals, and the distinction between daylight and darkness is used as an imperfect proxy for whether accidents occurred during the hours when the 100 km/h limit applied (Data Pipeline, Step 6). As a result, accidents occurring specifically between 06:00 and 19:00 cannot be isolated.

Second, the introduction of the speed limit coincided with the COVID-19 pandemic and the first lockdown. Although the comparison group, exposure correction and separate treatment of the transition years help account for this, COVID-19 may have affected treated and comparison roads differently, particularly due to differences in traffic patterns.

The comparability of the treated and comparison groups is another limitation. Comparison sections are often located near cities and may have more congestion and junctions than treated sections. Although road-section fixed effects account for permanent differences, the event-study indicates that pre-policy trends were not fully similar. This weakens the parallel-trends assumption required for a causal interpretation.

Another limitation concerns accident registration. BRON only includes registered accidents, and registration practices may change over time. For example, total registrations increased from approximately 104,000 in 2020 to 134,000 in 2023. Although the comparison-group design partly accounts for changes affecting both groups, differences in registration between road types may still influence the results.

The INWEVA exposure data also involve uncertainty. Approximately one-third of motorway traffic volumes are derived or modelled rather than directly measured. Furthermore, the estimation method changed in 2020 and again in 2024, potentially causing breaks in traffic-volume trends. Rijkswaterstaat reports that for approximately 90% of the network length, the new method differs by less than ±5% from the previous method (Rijkswaterstaat, 2025). These changes may therefore influence the estimated exposure and accident rates.

Furthermore, the lower speed limit may have influenced route choice, with some traffic potentially shifting from motorways to other roads (Rijkswaterstaat, 2021). Such shifts are partly reflected in motorway exposure data, but possible safety effects on other roads fall outside the scope of this study.

The analysis of accident severity is limited by the small number of fatal accidents. Fatal accidents are therefore combined with injury accidents, preventing a separate analysis at road-section level. In addition, actual driving speeds were not measured. The Rijkswaterstaat monitoring study is used only to provide context for possible changes in driving speeds.

Finally, despite the use of a comparison group, exposure correction and fixed effects, this remains an observational study. The results therefore indicate an association between the introduction of the speed limit and changes in traffic safety, rather than definitive proof of a causal effect.

8.3 Implications and future research

The results of this study have several implications for traffic safety policy. The findings suggest that the introduction of the 100 km/h daytime speed limit was associated with a

more favourable development in accident rates on treated motorway sections compared with comparison sections. This difference was particularly pronounced for injury and fatal accidents. Although the results cannot establish a definitive causal effect, they are consistent with the expectation that lower driving speeds can improve traffic safety, especially by reducing the risk of more severe accidents. The findings therefore provide support for considering speed limits as a potential instrument for improving motorway safety.

Future research could address several limitations of this study. First, access to accident data containing exact dates and times would allow researchers to distinguish directly between accidents occurring during daytime and nighttime. Second, using traffic exposure data separately for daylight and darkness would improve the comparison of accident risks under the two speed-limit regimes. Third, including actual driving-speed measurements would help establish whether and to what extent the lower speed limit changed driving behaviour. Finally, using more pre-policy years and selecting comparison sections with more similar traffic characteristics could improve the assessment of parallel trends and strengthen the reliability of the estimated effects.

Overall, further research with more detailed accident, traffic and speed data would help establish more accurately how the 100 km/h daytime speed limit affects traffic safety on Dutch motorways.

9. Conclusion

This study examined how traffic safety on Dutch motorways changed after the daytime speed limit was reduced from 120/130 km/h to 100 km/h in March 2020, compared with motorway sections where the limit did not change. Because the policy coincided with the COVID-19 pandemic and the open accident data contain no time of day, the analysis separated three explanations step by step: less traffic, nationwide trends, and a change specific to the roads and hours where the limit was lowered.

Registered accidents fell by 21.9% on treated sections, against 0.3% on comparison sections **(SQ1)**. This difference was not only due to less traffic: after accounting for vehicle-kilometres, accident rates on treated roads developed 21.7% more favourably than on comparison roads **(SQ2)**. This estimate remained similar for alternative before and after periods and a stricter comparison group (16.5% to 23.3%). It was largest on sections with high accident rates and not significant on sections with the lowest rates.

The relative improvement was larger for severe accidents: 35.0% for injury and fatal accidents, against 20.7% for material-damage-only accidents. This is consistent with the Power Model, although injury and fatal accident rates rose in both groups, so the improvement is relative rather than absolute. No single crash type explains the change **(SQ3)**. The relative change was also 11.8% more favourable during daylight, which largely overlaps with the hours of the 100 km/h limit, than during darkness **(SQ4)**.

In answer to the main research question, traffic safety developed more favourably on motorway sections where the daytime limit was lowered than on sections where it did not change. Accident rates developed 21.7% more favourably, injury and fatal accident rates 35.0% more favourably, and the relative change was 11.8% more favourable in daylight than

in darkness. These findings consistently point in the direction expected from a lower speed limit. However, because the two groups already differed significantly in 2018 and light condition is only an imperfect proxy for the hours of the limit, they indicate an association rather than proof of a causal effect. More pre-policy years and light-specific traffic data would be needed to establish the size of the effect with more certainty.

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Appendix A: Author contribution statement

All group members jointly contributed to selecting the research topic and formulating the research questions. The data collection was divided among the group members. Data processing, analysis and the final review of the report were carried out collaboratively. The writing of the report was divided among the group members, with each member taking responsibility for specific sections.

Group members and sections:

- Roos: was responsible for writing Chapter 1 (Introduction) and Chapter 6 (Results). Also contributed to the development and writing of the project proposal with Harmen. Responsible for interpreting and describing the results of all four sub-questions and presenting the findings in a clear manner.
- Meike was responsible for writing Chapter 5 (Analytical Methods) and Chapter 8 (Discussion).
- Harmen: was responsible for the data pipeline: wrote the script that downloads BRON and INWEVA, linked accidents to INWEVA road sections and computed the vehicle-kilometres. Also built the analyses for sub-questions 2–4 and wrote the project proposal, from which Chapters 1–4 are adapted.
- Teun: Contributed to the datapipeline, especially for sub-research questions 1 and 2 and the robustness check. Played a supportive role in contribution to the pipeline used for the results in sub-research question 3 and 4.
- Lieke: was responsible for the geographical scale and final classification in Chapter 3, including the map of treated and comparison motorway sections and writing Chapter 9.

Appendix B: AI statement

We acknowledge the use of ChatGPT (OpenAI, https://chatgpt.com) and Claude (Anthropic, https://claude.ai) in preparing this report. These generative AI tools were used as supportive aids and were not used to generate the research or the report as a whole. The group remained responsible for the research choices, analysis, interpretation of the results, conclusions and final editing.

Purposes of AI use

- **Translation and language refinement:** translating Dutch text into English and improving the grammar, clarity, and academic wording of the report.
- **Programming assistance:** assisting with programming, checking, and debugging Python code, clarifying code comments, and improving the readability and structure of the code.
- **Sparring:** verifying the group's own interpretation of figures and results and its formulation of conclusions, and helping group members understand how the code and statistical models work.
- **Report structure and review:** suggesting improvements to the layout, structure, and presentation of the report, and reviewing the report against the grading rubric.

Use per group member

- **Roos:** I used ChatGPT for Chapter 6 (Results), to translate Dutch text into English, act as a sparring partner to verify my own interpretation of the figures, and suggest improvements to the layout and presentation of the chapter.
- **Meike:** I used ChatGPT for Chapter 5 (Analytical Methods), Chapter 6 (Results), and Chapter 8 (Discussion), to translate Dutch text into English, act as a sparring partner to verify my own interpretation of figures and results and my formulation of conclusions, and improve clarity, structure, and academic wording.
- **Harmen:** I used Claude for the data pipeline and the analyses for sub-questions 2–4, to assist with programming and debugging in Python, clarify the comments in the code, and act as a sparring partner to help me understand how the code and statistical models work. I also used Claude to review the report against the grading rubric.
- **Teun:** I used ChatGPT for programming assistance (layout in plots and figures) and debugging in Python for subquestions SQ1 & SQ2 and as a sparring partner to challenge my conclusions for results in Ch7.
- **Lieke:** I used Claude for the midterm presentation, the maps of the final classifications, and Chapter 9 (Conclusion), to write and debug Python code for the maps, check English grammar, and review the report against the grading rubric.

Verification and academic integrity

All AI output was critically reviewed before it was included in the report or the underlying work. Suggestions were adapted, rewritten, or rejected where they were incorrect, too general, or did not fit the aim of the research. In particular:

- AI-generated or AI-suggested code was checked and tested by the group member concerned, and was only used once it worked correctly within the analysis.
- Interpretations and conclusions were formed by the group members themselves. AI was only used to verify them, and its feedback was assessed against the group's own data and results.
- Translated and edited text was checked to ensure that it accurately reflected the intended meaning.
- No AI-generated references were included without manual verification of their existence, relevance, and content.

The submitted work reflects our own understanding, and we take full responsibility for the accuracy, interpretation, and content of the final report.